Subhranil Deb

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Université Paris Saclay, Orsay

Education

Institut Mathématiques de Jussieu - Paris Rive Gauche — October 2026 - present
PhD in Mathematics, Double Kazhdan - Lusztig theory.
Advisors: Alexis Bouthier, Éric Vasserot

Université Paris Saclay — September 2025 - July 2026
M2 Analysis, Arithmetic, Geometry(AAG).
Advisor: Olivier Schiffmann

Chennai Mathematical Institute — August 2024 - May 2025
M1 Mathematics.

Chennai Mathematical Institute — August 2021 - June 2024
Bachelor of Science (Honours) in Mathematics and Computer Science.

Research Interests

Geometric Langlands Program, Geometric Representation Theory, Algebraic Geometry

Projects/Presentations

- **M2 Thesis** — October 2025 - Present
  Instructor: Olivier Schiffmann - Université Paris Saclay
  - I am currently reading about cohomological integrality for symmetric quotient stacks and the cohomological Mackey formula following the papers by Lucien Hennecart.
- **Geometric Satake Equivalence** — May 2025 - July 2025
  Instructor: Simon Riche - Université Clermont Auvergne
  - I read the theory of perverse sheaves, equivariant derived categories leading up to the Geometric Satake Equivalence. The main reference I followed is the book "Perverse Sheaves and Applications to Representation Theory" by Achar.
- **Degree Bounds For Syzygies of Linearly Reductive Groups** — May 2024
  Instructor: Manoj Kummini - Chennai Mathematical Institute
  - As a part of the course **Commutative Algebra I**, I presented "**Degree Bounds For Syzygies Of Invariants**" by H. Dersken. Here is the report.
- **Higher Dimensional Class Field Theory** — May 2024
  Instructor: Sukhendu Mehrotra - Chennai Mathematical Institute
  - As a part of the course **Geometric Class Field Theory**, I presented "**Non-abelian class field theory for arithmetic surfaces**" by Hoffman, Walter and Wiesend. Here is the report.
- **Construction of Hilbert and Quot Schemes** — May 2024
  Instructor: Mohit Upmanyu and Bivas Khan - Chennai Mathematical Institute
  - As a part of the course **Algebraic Geometry II**, I presented the construction of Hilbert and Quot Schemes using Castelnuovo-Mumford regularity. Here is the report.
- **Iwasawa Theory of Taelman Class Modules** — Dec 2023
  Instructor: Anwesh Ray - Chennai Mathematical Institute
  - I read the book Drinfeld Modules by Mihran Papikian for my course on Function Field Arithmetic. As a part of the course **Function Field Arithmetic**, I presented Higgins thesis: "**Iwasawa theory of Taelman class modules**". Here is the report.
- **Serre Duality on Compact Riemann Surfaces and Algebraic Curves** — May 2023 - June 2023
  Instructor: Yogish Holla - TIFR, Mumbai
  - I read R.R. Simha's paper "**The Riemann–Roch Theorem for Compact Riemann Surfaces**" for a sheaf theoretic proof of Serre Duality.
  - As a part of my final presentation, I read the first 3 chapters of Diamond-Shurman's text on modular forms and presented the dimension formula as an application of the Riemann–Roch Formula.

Talks

Perverse Sheaves and The Geometric Satake Equivalence, CMI Student Seminars(website), August 2025

BGG Resolution and Higher Ext Groups in Category $\mathcal{O}$, Lie Algebra Representations Course, June 2025

Quasicoherent Sheaves on Toric Varieties, Toric Varieties Course, April 2025

Grothendieck local duality, Commutative Algebra II Course, April 2025

Clifford Algebras, Pin and Spin Groups, Lie Groups and Lie Algebra Course, Nov 2024

Galois Representations and Iwasawa Theory, CMI Student Seminars(website), Jan 2024

Modular forms and Riemann-Roch Theorem, VSRP presentation, July 2023

Achievements

2025	**Awarded the Sophie Germain Scholarship for admission in the M2 AAG program by FMJH,** Université Paris Saclay
2024	**Selected for a fully funded research internship by ReLaX under Prof. Simon RICHE,** Université Clermont Auvergne
2024	**Awarded Scholarship for admission in the MSc Mathematics program at CMI,** Chennai Mathematical Institute
2023	**Selected for TIFR VSRP (Visiting Students' Research Program),** Tata Institute of Fundamental Research (TIFR), Mumbai
2022	**Selected for Madhava Nurture camp,** S.P. College & HBCSE
2019	**Qualified Regional Mathematics Olympiad(RMO),** HBCSE
2019	**Qualified MTRP,** Indian Statistical Institute, Kolkata

Conferences/Workshops

- Derived Algebraic Geometry, CoHAs and Operads — June 2026
  Université Angers
- Enumerative geometry, rationality, derived categories and interactions — June 2026
  Université Paris Saclay
- Mathematics Inspired by Physics — June 2026
  IHES
- CARE - Collaborations in Algebra, Representation Theory and Ethics — October 2025
  ENS Lyon
- Geometry and Representation Theory around the Langlands Programs — July 2025
  Université Clermont Auvergne
- Introduction to the p-adic local Langlands program — September 2024
  Indian Institute of Technology Jammu
- Derived categories and semi-orthogonal decompositions — June 2024
  Chennai Mathematical Institute
- Rational Points on Modular Curves — September 2023
  International Centre of Theoretical Sciences(Online)

Teaching Experience

I have been a teaching assistant for the following courses at CMI:

Topology (Instructor: Prof. Aditya Karnataki)
Algebra I (Instructor: Prof. Aditya Karnataki)
Topology (Instructor: Prof. Parameswaran Sankaran)
Analysis 3 (Instructor: Prof. Srinivasan Vasanth)
Analysis 2 (Instructor: Prof. Upendra Kulkarni)
Advanced Programming in Python (Instructor: Prof. Samir Datta)
Algebra 1 (Instructor: Prof. Upendra Kulkarni)
Introduction To Programming In Haskell (Instructor: Prof. S P Suresh)